Cap 3 - Logaritmos

Exemplos

Ex 3.1 a) $\log_2 16 \Rightarrow 2^x = 16$

$2^x = 2^4 \Rightarrow x = 4$

16	2
8	2
4	2
2	2

$$\log_2 16 = 4$$

b) $\log_5 \frac{1}{25} \Rightarrow 5^x = 25^{-1}$

$5^x = (5^2)^{-1} \Rightarrow x = -2$

25	5
5	5

$$\log_5 \frac{1}{25} = -2$$

c) $\log_7 1 \Rightarrow 7^x = 1 \Rightarrow 7^x = 7^0 \Rightarrow x = 0$

$$\log_7 1 = 0$$

d) $\log_5 \sqrt[3]{5} \Rightarrow 5^x = \sqrt[3]{5}$

$5^x = 5^{1/3} \Rightarrow x = 1/3$

$$\log_5 \sqrt[3]{5} = \frac{1}{3}$$

Ex 3.2 Devemos ter: $(\log_{\sqrt{3}}(x-5))$

(i) $b > 0 \Rightarrow b = 3$ ✓

(ii) $b \neq 1 \Rightarrow b = 3$ ✓

(iii) $a > 0 \Rightarrow x - 5 > 0 \Rightarrow x > 5$

$$\boxed{\log_3(x-5) \ \exists \forall x \in \mathbb{R} / x > 5}$$

Ex 3.3 $\log \frac{1}{1000} \Rightarrow 10^x = 1000^{-1}$

$10^x = 10^{-3} \Rightarrow x = -3$

$$\boxed{\log \frac{1}{1000} = -3}$$

Ex 3.4 $\log 1,4$; $2 = 10^{0,301}$; $7 = 10^{0,845}$

$\log 1,4 \Rightarrow 10^x = 1,4 \Rightarrow 10^x = \frac{14}{10} \Rightarrow 10^x = \frac{2.7}{10}$

$10^x = \frac{10^{0,301} \cdot 10^{0,845}}{10^1} \Rightarrow 10^x = \frac{10^{1,146}}{10^1} =$

$10^x = 10^{0,146} \Rightarrow x = 0,146$

$$\boxed{\log 1,4 = 0,146}$$

Ex 3.5 a) $\log_2(4 \cdot 2) = \log_2 4 + \log_2 2$

$\log_2 8 = \log_2 4 + \log_2 2$

$3 = 2 + 1 = 3$

b) $\log_5(625 \cdot 125) = \log_5 625 + \log_5 125$

$\log_5 78125 = \log_5 625 + \log_5 125$

78125 | 5
15625 | 5
3125 | 5
625 | 5
125 | 5
25 | 5
5 | 5

} 7

625 | 5
125 | 5
25 | 5
5 | 5

} 4

125 | 5
25 | 5
5 | 5

} 3

$7 = 4 + 3 = 7$

Ex 3.6 a) $\log_2\left(\frac{8}{2}\right) = \log_2 8 - \log_2 2$

$\log_2 4 = \log_2 8 - \log_2 2$

$2 = 3 - 1 = 2$

b) $\log_5 \frac{625}{125} = \log_5 625 - \log_5 125$

$1 = 4 - 3 = 1$

Ex 3.7 a) $\log 7,2 \Rightarrow \log \frac{72}{10} \Rightarrow \log 72 - \log 10$

$\log (2 \cdot 2 \cdot 2 \cdot 3 \cdot 3) - \log 10 \Rightarrow \log (2^3 3^2) - \log 10$

$\log 2^3 + \log 3^2 - \log 10 \Rightarrow 3 \log 2 + 2 \log 3 - \log 10$

Agora, usando os dados do problema

$3 \cdot 0,301 + 2 \cdot 0,477 - 1 \Rightarrow \boxed{\log 7,2 = 0,857}$

b) $\log \sqrt{3} \Rightarrow \log 3^{1/2} \Rightarrow \frac{1}{2} \log 3 \Rightarrow \frac{1}{2} \cdot 0,477$

$$\boxed{\log \sqrt{3} = 0,2385}$$

Ex 3.8 a) $\log_x (3x^2 - x) = 2 \Rightarrow x^2 = 3x^2 - x$

$2x^2 - x = 0$

$x(2x - 1) = 0 \Rightarrow \begin{cases} x' = 0 \\ x'' = 1/2 \end{cases}$

Como a base deve ser $\neq 0$, então

$$\boxed{S = \{1/2\}}$$

b) $(\log_3 x)^2 - \log_3 x - 6 = 0$

Aqui, vamos fazer uma mudança de variável:

$\log_3 x = y$. Então:

$$y^2 - y - 6 = 0$$

$$\begin{cases} a = 1 \\ b = -1 \\ c = -6 \end{cases} \qquad \frac{1 \pm \sqrt{1 + 24}}{2} \Rightarrow \begin{matrix} y' = 3 \\ y'' = -2 \end{matrix}$$

Agora, determinando x $\Rightarrow$ $\log_3 x = 3 \Rightarrow x = 27$

$\log_3 x = -2 \Rightarrow x = 1/9$

Logo:

$$\boxed{S = \left\{\frac{1}{9}, 27\right\}}$$

c) $\log_5 (\log_3 x) = 1 \Rightarrow$

$\log_3 x = 5$

$3^5 = x$

$x = 243$

$$\boxed{S = \{243\}}$$

Ex 3.9 $h(t) = 1,5 + \log_3(t+1)$

$3,5 - 1,5 = \log_3(t+1)$

$2 = \log_3(t+1)$

$t+1 = 3^2$

$t+1 = 9 \Rightarrow$ **$t = 8$ anos**

Ex 3.10 $\log_2 6$; $\log 2 = 0,3$; $\log 3 = 0,4$

$\log_2(2 \cdot 3) = \log_2 2 + \log_2 3$

$= 1 + \frac{\log 3}{\log 2} = 1 + \frac{0,4}{0,3}$

$\log_2 6 = 2,33\ldots$

Ex 3.11 $\log_2 x + \log_4 x + \log_{16} x = 7$

Agora, escreveremos todos os logs na base 2

$\log_2 x + \frac{\log_2 x}{\log_2 4} + \frac{\log_2 x}{\log_2 16} = 7$

$\log_2 x + \frac{\log_2 x}{2} + \frac{\log_2 x}{4} = 7$

07

$$\frac{4\log_2 x + 2\log_2 x + \log_2 x}{4} = 7$$

$$\log_2 x \cdot (4+2+1) = 4 \cdot 7$$

$$\log_2 x = 4 \Rightarrow 2^4 = x \Rightarrow 16 = x$$

$\log_2$:

$$S = \{16\}$$

Lista de Exercícios

Logaritmos

(01) a) $3^x = 5 \Rightarrow \log_3 5 = x$

$\log_3 5 = \frac{\log 5}{\log 3} = \frac{0,699}{0,477} =$ **$x = 1,465$**

b) $10^x = 21 \Rightarrow \log 21 = x \Rightarrow$ **$x = 1,322$**

c) $10^x = 15 \Rightarrow \log 15 = x \Rightarrow \log 3.5$

$\log 3 + \log 5 \Rightarrow 0,477 + 0,699 \Rightarrow$ **$x = 1176$**

d) $2^x = 42 \Rightarrow \log_2 42 = x$

$\log_2 2.21 \Rightarrow \log_2 2 + \log_2 21$

$1 + \frac{\log 21}{\log 2} \Rightarrow 1 + \frac{1,322}{0,301}$

$x = 5,392$

(02)

(02) a) $\log_{\sqrt{8}} 4 = x \Rightarrow \sqrt{8}^{x} = 4$

$8^{x/2} = 4$

$2^{\frac{3x}{2}} = 2^2 \Rightarrow \frac{3x}{2} = 2$

$$x = \frac{4}{3}$$

b) $\log_{25} 0{,}2 \Rightarrow 25^{x} = 0{,}2 \Rightarrow 25^{x} = \frac{2}{10} = \frac{1}{5}$

$5^{2x} = 5^{-1} \Rightarrow 2x = -1 \Rightarrow$

$$x = -\frac{1}{2}$$

c) $\log_2 \sqrt[3]{64} \Rightarrow 2^{x} = 64^{1/3}$

$2^{x} = 2^{6/3}$

$2^{x} = 2^2$

$$x = 2$$

64	2
32	2
16	2
8	2
4	2
2	2

d) $\log_{16} 32 \Rightarrow 16^{x} = 32$

$2^{4x} = 2^5$

$4x = 5 \Rightarrow$

$$x = \frac{5}{4}$$

e) $\log_5 0{,}000064 \Rightarrow 5^x = \frac{64}{1000000} = \frac{1}{15625}$

$5^x = \frac{1}{5^6} \Rightarrow 5^x = 5^{-6}$

$\boxed{x = -6}$

f) $\log_{49} \sqrt[3]{7} \Rightarrow 49^x = 7^{1/3}$

$7^{2x} = 7^{1/3}$

$2x = \frac{1}{3} \Rightarrow \boxed{x = \frac{1}{6}}$

g) $\log_3 81 \Rightarrow 3^x = 81 \Rightarrow 3^x = 3^4$

$\boxed{x = 4}$

h) $\log_2 \sqrt[8]{64} \Rightarrow 2^x = 64^{1/8}$

$2^x = 2^{6/8}$

$\boxed{x = \frac{3}{4}}$

(69)

i) $\log_4 2\sqrt{2} \Rightarrow 4^x = 2\sqrt{2}$

$2^{2x} = 2\,2^{1/2}$

$2^{2x} = 2^{3/2}$

$2x = \frac{3}{2} \Rightarrow \boxed{x = \frac{3}{4}}$

ı) $\log_2 0{,}25 \Rightarrow 2^x = \frac{25}{100} = \frac{1}{4} \Rightarrow 2^x = 4^{-1}$

$2^x = 2^{-2} \Rightarrow \boxed{x = -2}$

k) $\log_{\sqrt[5]{2}} 128 \Rightarrow 2^{\frac{1}{5}x} = 128 \Rightarrow 2^{\frac{x}{5}} = 2^7$

$\frac{x}{5} = 7 \Rightarrow \boxed{x = 35}$

l) $\log_{625} \sqrt{5} \Rightarrow 625^x = 5^{1/2}$

$5^{4x} = 5^{1/2}$

$4x = \frac{1}{2} \Rightarrow \boxed{x = \frac{1}{8}}$

03 a) $\log 36 \Rightarrow \log(2.3.2.3) \Rightarrow \log 2^2.3^2$

$2\log 2 + 2\log 3$

$2.0{,}301 + 2.0{,}477 \Rightarrow \boxed{\log 36 = 1{,}556}$

b) $\log 30 \Rightarrow \log 3.2.5 \Rightarrow \log 3 + \log 2 + \log 5$

$0{,}477 + 0{,}699 + 0{,}301 \Rightarrow \boxed{\log 30 = 1{,}477}$

c) $\log 180 \Rightarrow \log 5.2.3.2.3 \Rightarrow$

$\log 5 + 2\log 2 + 2\log 3 \Rightarrow \boxed{\log 180 = 2{,}255}$

04 a) $\underbrace{\log_{10} 0{,}001}_{*} + \underbrace{\log_3 3\sqrt{3}}_{**} - \underbrace{\log_2 16}_{***}$

$* \quad 10^x = 0{,}001 \Rightarrow 10^x = 10^{-3} \Rightarrow x = -3$

$** \quad 3^x = 3.3^{1/2} \Rightarrow 3^x = 3^{\frac{3}{2}} \Rightarrow x = 3/2$

$*** \quad 8^x = 16 \Rightarrow 2^{3x} = 2^4 \Rightarrow x = \frac{4}{3}$

$-3 + 3/2 - 4/3 \Rightarrow \boxed{-17/6}$

06

b) $\underbrace{\log_{1/2} 8}_{*} - \underbrace{\log_{4/3} \frac{27}{64}}_{**} + \underbrace{\log_2 1024}_{***}$

* $\left(\frac{1}{2}\right)^x = 8 \Rightarrow 2^{-x} = 2^3 \Rightarrow x = -3$

** $\left(\frac{4}{3}\right)^x = \frac{27}{64} \Rightarrow \frac{3^{-x}}{4^{-x}} = \frac{3^3}{4^3} \Rightarrow x = 3$

*** $2^x = 1024 \Rightarrow 2^x = 2^{10} \Rightarrow x = 10$

$-3 - (-3) + 10 = \boxed{10}$

05 a) $\log_2 (x-8) \Rightarrow x - 8 > 0$

$x > 8$

$\boxed{S = \{x \in \mathbb{R} / x > 8\}}$

b) $\log (1-x) \Rightarrow 1 - x > 0$

$x < 1$

$\boxed{S = \{x \in \mathbb{R} / x < 1\}}$

c) $\log_5(5x-2) + \log_5(x-3)$

Aqui temos duas condições. Deve-se encontrar a interseção:

$5x - 2 > 0$

$x > \frac{2}{5}$

$x - 3 > 0$

$x > 3$

A condição $x > 3$ satisfaz $x > 2/5$, logo

$$\boxed{S = \{x \in \mathbb{R} / x > 3\}}$$

(06) $$\frac{\log_2 1 + \log_{10} 0{,}01}{\log_2 \frac{1}{64} \cdot \log_4 \sqrt{8}} = \frac{0 + (-2)}{-6 \cdot 3/4}$$

$\log_4 \sqrt{8}$

$\log 4^x = 8^{1/2}$

$2^{2x} = 2^{3/2}$

$x = 3/4$

$$\frac{-2}{-9/2} = \boxed{\frac{4}{9}}$$

07) a) $\log_9 1 = 0$

b) $\log_{\frac{1}{7}} 1 = 0$

c) $\log_{33} 1 = 0$

d) $\log_8 8 = 1$

e) $\log_{13} 13 = 1$

f) $\log_4 4^6 = 6 \cdot \log_4 4^1 = \log_4 4^6 = 6$

g) $\log_5 5^{-7} = -7 \cdot \log_5 5^1 = \log_5 5^{-7} = -7$

h) $3^{\log_3 27} = 27$

i) $4^{\log_4 \frac{1}{2}} = \frac{1}{2}$

j) $\log_4 4 = 1$

k) $\log_{\frac{1}{7}} \frac{1}{7} = 1$

l) $\log_6 1 = 0$

m) $\log_{0,2} 1 = 0$

n) $\log_6 6^2 = 2 \log_6 6 - \log_6 6^2 = 2$

o) $\log_{\frac{1}{10}} \left(\frac{1}{10}\right)^3 = 3 \cdot \log_{\frac{1}{10}} \frac{1}{10} = \log_{\frac{1}{10}} \left(\frac{1}{10}\right)^3 = 3$

p) $5^{\log_5 7} = 7$

q) $\log_4 4^3 = 3 \log_4 4 - \log_4 4^3 = 3$

08) a) $\log 15 = \log 3.5 = \log 3 + \log 5$

$0,477 + 0,699 =$

$\boxed{\log 15 = 1,176}$

b) $\log 14 = \log 2 + \log 7 = 0,301 + 0,845$

$\boxed{\log 14 = 1,146}$

c) $\log 42 = \log 2.3.7 = \log 2 . \log 3 + \log 7$

$0,301 + 0,477 + 0,845 =$ $\boxed{\log 42 = 1,623}$

d) $\log 210 = \log 3.7.2.5 =$

$0,301 + 0,477 + 0,699 + 0,845 =$ $\boxed{\log 210 = 2,322}$

e) $\log \frac{2}{3} = \log 2 - \log 3 = 0,301 - 0,477$

$\boxed{\log \frac{2}{3} = -0,176}$

f) $\log 1,5 = \log \frac{15}{10} = \log 3 + \log 5 - \log 2 - \log 5$

$\log 3 - \log 2$

$0,477 - 0,301$ $= \boxed{\log 1,5 = 0,176}$

g) $\log 0,6 = \log \frac{6}{10} = \log 2 + \log 3 - \log 2 - \log 5$

$\log 3 - \log 5$
$0,477 - 0,699$ $\Rightarrow$ $\boxed{\log 0,6 = -0,222}$

h) $\log 1,2 = \log \frac{12}{10} = \log 2 + \log 2 + \log 3 - \log 2 - \log 5$

$\log 2 + \log 3 - \log 5$
$0,301 + 0,477 - 0,699$ $\Rightarrow$ $\boxed{\log 1,2 = 0,079}$

i) $\log 8 = \log 2^3 = 3 \log 2 = 3 \cdot 0,301$

$\boxed{\log 8 = 0,903}$

o) $\log 81 = \log 3^4 = 4 \log 3 = 4 \cdot 0,477$

$\boxed{\log 81 = 1,908}$

K) $\log \sqrt{2} = \log 2^{1/2} = \frac{1}{2} \log 2 = \frac{1}{2} \cdot 0,301$

$\boxed{\log \sqrt{2} = 0,1505}$

(12)

l) $\log \sqrt[3]{25} = \log (5^2)^{\frac{1}{3}} = \frac{2}{3} \log 5$

$\frac{2}{3} \cdot 0,699 =$ $\boxed{\log \sqrt[3]{25} = 0,466}$

m) $\log \sqrt[5]{6} = \log (2 \cdot 3)^{\frac{1}{5}} = \frac{1}{5} (\log 2 + \log 3)$

$\frac{1}{5} (0,301 + 0,477) =$ $\boxed{\log \sqrt[5]{6} = 0,1556}$

n) $\log 1,8 = \log \frac{18}{10} = \log 3 + \log 3 + \log 2 - \log 2 - \log 5$

$2 \cdot 0,477 - 0,699 =$ $\boxed{\log 1,8 = 0,255}$

o) $\log \frac{1}{27} = \log 1 - 3 \log 3 = 0 - 3 \cdot 0,477$

$\boxed{\log \frac{1}{27} = -1,431}$

p) $\log \frac{4}{25} = 2\log 2 - 2\log 5 \Rightarrow 2(0,301 - 0,699)$

$$\boxed{\log \frac{4}{25} \div -0,796}$$

q) $\log_2 50 = \frac{\log 50}{\log 2} = \frac{\log 2 + 2\log 5}{\log 2}$

$$\frac{0,301 + 2 \cdot 0,699}{0,301} \Rightarrow \boxed{\log_2 50 = 5,645}$$

r) $\log_3 45 = \frac{\log 45}{\log 3} \Rightarrow \frac{2 \cdot \log 3 + \log 5}{\log 3}$

$$\frac{2 \cdot 0,477 + 0,699}{0,477} \Rightarrow \boxed{\log_3 45 = 3,465}$$

s) $\log_9 2 \Rightarrow \frac{\log 2}{\log 9} \Rightarrow \frac{\log 2}{2\log 3}$

$$\frac{0,301}{2 \cdot 0,477} \Rightarrow \boxed{\log_9 2 = 0,316}$$

(14)

e) $\log_8 600 = \frac{\log 600}{\log 8} =$

$$\frac{\log 2 + \log 3 + \log 2 + \log 5 + \log 2 + \log 5}{3 \cdot \log 2}$$

$$\frac{3 \cdot (0,301) + 0,477 + 2 \cdot 0,699}{3 \cdot 0,301}$$

$$\boxed{\log_8 600 = 3,076}$$

a) $\log_5 3 = \frac{\log 3}{\log 5} = \frac{0,477}{0,699} =$ $\boxed{\log_5 3 = 0,682}$

v) $\log_6 15 = \frac{\log 15}{\log 6} = \frac{\log 3 + \log 5}{\log 3 + \log 2}$

$$\frac{0,477 + 0,699}{0,477 + 0,301} =$$

$$\boxed{\log_6 15 = 1,612}$$

09) a) $\log 5 + \log 200 = \log 1000 \Rightarrow 10^5 = 1000$

$\boxed{x = 3}$

b) $\log 100 + \log 50 + \log 10 + \log 2 = \log 100000$

$10^x = 100000 \Rightarrow \boxed{x = 5}$

c) $\log_2 24 - \log_2 3 = \log_2 \left(\frac{24}{3}\right) = \log_2 8$

$2^x = 8 \Rightarrow \boxed{x = 3}$

d) $\log_5 8 + \log_5 12,5 - \log_5 4 = \log_5 \left(\frac{8 \cdot 12,5}{4}\right)$

$\log_5 \frac{100}{4} \Rightarrow \log_5 25 \Rightarrow \boxed{x = 2}$

e) Aqui necessita-se mudança de base (convenientemente)

$\log_3 2 \cdot \log_2 5 \cdot \log_5 3 = \frac{\log_2 2}{\log_2 3} \cdot \log_2 5 \cdot \frac{\log_2 3}{\log_2 5}$

$\boxed{\log_2 2 = 1}$

(16)

f) $\log_3 5 \cdot \log_4 27 \cdot \log_{25} \sqrt[3]{2}$

$\log_3 5 \cdot \log_4 3^3 \cdot \log_{25} 2^{1/3}$

$3 \cdot \frac{1}{3} \cdot \log_3 5 \cdot \log_4 3 \cdot \log_{25} 2$

$\log_3 5 \cdot \log_4 3 \cdot \frac{\log_3 2}{2\log_3 5} \Rightarrow \frac{1}{2} \log_4 3 \cdot \log_3 2$

$\frac{1}{2} \log_4 3 \cdot \frac{\log_4 2}{\log_4 3} \Rightarrow \frac{1}{2} \cdot \log_4 2$

$\frac{1}{2} \cdot \left[4^x = 2 \right] \Rightarrow \frac{1}{2} \left(2^{2x} = 2^1 \right) = \frac{1}{2} \cdot \frac{1}{2}$

$$\boxed{x = \frac{1}{4}}$$

(10) a) $\log_2 |x-1| + 3 = \log_2 |7x+4|$

Condições de existência

$x - 1 > 0$ $\qquad$ $7x + 4 > 0$

$x > 1$ $\qquad$ $x > -\frac{4}{7}$

Logo, $\boxed{x > 1}$

Então: $\log_2(x-1) - \log_2(7x+4) = -3$

$$\log_2\left(\frac{x-1}{7x+4}\right) = -3$$

$$\frac{x-1}{7x+4} = 2^{-3} \Rightarrow \frac{x-1}{7x+4} = \frac{1}{8}$$

$$8x - 8 = 7x + 4 \Rightarrow x = 12$$

$$\boxed{S = \{x \in \mathbb{R} / x = 12\}}$$

b) $\log(x-5) + \log(2x-20) = 1 + \log(3x-35)$

C.E: $x - 5 > 0 \quad 2x - 20 > 0 \quad 3x - 35 > 0$

$x > 5 \quad x > 10 \quad x > \frac{35}{3} \simeq 11{,}67$

$$\log\frac{(x-5)\cdot(2x-20)}{3x-35} = 1 \Rightarrow \frac{2x^2 - 30x + 100}{3x-35} = 10$$

$$2x^2 - 30x + 100 = 30x - 350 \Rightarrow 2x^2 - 60x + 450 = 0$$

$$x^2 - 30x + 225 = 0$$

$\begin{cases} a = 1 \\ b = -30 \\ c = 225 \end{cases}$ $\quad \frac{30 \pm \sqrt{900 - 900}}{2} \Rightarrow x = 15$

$$\boxed{S = \{x \in \mathbb{R} / x = 15\}}$$

(18)

C.E.

c) $2\log x = \log 4 + \log 3x \qquad (x>0)$

$2\log x = \log 12x \Rightarrow 2\log x = \log 12 + \log x$

$2\log x - \log x = \log 12$

$\log x = \log 12 \Rightarrow x = 12$

$$S = \{x \in \mathbb{R} / x = 12\}$$

d) $\log(x+4) + \log(x-4) = 2\log 3$

$\log(x^2 - 16) = 2\log 3$

$\log(x^2 - 16) = \log 3^2$

$\log\left(\frac{x^2-16}{9}\right) = 10^0 \Rightarrow x^2 - 16 = 9 \Rightarrow x = \pm\sqrt{25}$

$x = \pm 5$

Condições de existência: $x+4>0$ $\qquad$ $x-4>0$

$x > -4$ $\qquad$ $x > 4$

Logo:

$$S = \{x \in \mathbb{R} / x > 5\}$$

C.E.

e) $\log_5 x + \log_{25} x = 3 \qquad (x > 0)$

$\log_5 x + \frac{\log_5 x}{\log_5 25} = 3$

$\log_5 x + \frac{\log_5 x}{2} = 3 \Rightarrow \frac{3}{2}\log_5 x = 3$

$\log_5 x = 2 \Rightarrow x = 25$

$S = \{x \in \mathbb{R} / x = 25\}$

f) $\log_x 4 + \log_2 x = 3 \qquad (x > 0 \text{ e } x \neq 1)$

$\frac{\log_2 4}{\log_2 x} + \log_2 x = 3 \Rightarrow 2 + (\log_2 x)^2 - 3\log_2 x = 0$

Chamando $\log_2 x$ de "y" $\Rightarrow y^2 - 3y + 2 = 0$

$\begin{cases} a = 1 \\ b = -3 \\ c = 2 \end{cases} \quad \frac{3 \pm \sqrt{9 - 8}}{2} \Rightarrow \begin{matrix} y' = 2 \\ y'' = 1 \end{matrix}$

Agora, temos: $\log_2 x = 2 \qquad \log_2 x = 1$

$x = 4 \qquad x = 2$

$S = \{2, 4\}$

(20)

g) $\log_x 243 = 5$

$x^5 = 243$

$3^5 = 243$

$\boxed{x = 3}$

243	3
81	3
27	3
9	3
3	3

h) $\log_x \frac{1}{9} = 2 \Rightarrow x^2 = \frac{1}{9} \Rightarrow x = \pm\sqrt{\frac{1}{9}} = \pm\frac{1}{3}$

Como a base $x > 0$, então:

$\boxed{x = \frac{1}{3}}$

i) $\log_3 \frac{x+3}{x-1} = 1 \Rightarrow \frac{x+3}{x-1} = 3$

$x + 3 = 3x - 3 \Rightarrow 2x = 6 \Rightarrow \boxed{x = 3}$

j) $\log_{\frac{1}{3}} (x-1) = -2 \Rightarrow \left(\frac{1}{3}\right)^{-2} = (x-1)$

$9 = x - 1 \Rightarrow \boxed{x = 10}$

k) $\log_5(1-4x) = 2 \Rightarrow 5^2 = 1-4x$

$4x = -24 \Rightarrow \boxed{x = -6}$

l) $\log_{12}(x^2 - x) = 1 \Rightarrow x^2 - x - 12 = 0$

$\begin{cases} a = 1 \\ b = -1 \\ c = -12 \end{cases}$ $\quad \frac{1 \pm \sqrt{1+48}}{2} \Rightarrow \begin{matrix} x' = 4 \\ x'' = -3 \end{matrix}$

$\boxed{S = \{-3, 4\}}$

m) $\log_{\frac{1}{2}}(x^2 + 4x - 5) = -4 \Rightarrow x^2 + 4x - 5 = 16$

$x^2 + 4x - 21 = 0$

$\begin{cases} a = 1 \\ b = 4 \\ c = -21 \end{cases}$ $\quad \frac{-4 \pm \sqrt{16+84}}{2} \Rightarrow \frac{-4 \pm 10}{2} \Rightarrow \begin{matrix} x' = -7 \\ x = 3 \end{matrix}$

$\boxed{S = \{-7, 3\}}$

n) $\log_{x+2}(20 - 2x) = 2 \Rightarrow (x+2)^2 = 20 - 2x$

$x^2 + 4x + 4 - 20 + 2x = 0$

$x^2 + 6x - 16 = 0$

$\begin{cases} a = 1 \\ b = 6 \\ c = -16 \end{cases}$ $\quad \frac{-6 \pm \sqrt{36+64}}{2} \Rightarrow \frac{-6 \pm 10}{2} \Rightarrow \begin{matrix} x' = -8 \\ x'' = 2 \end{matrix}$

$\boxed{S = \{-8, 2\}}$

(22)

o) $\log_{x-1}(1-2x) = 2$

$(x-1)^2 = 1-2x \Rightarrow x^2 - 2x + 1 = 1 - 2x$

$x^2 = 0 \Rightarrow \boxed{x = 0}$

Obs.: O resultado é válido somente porque está na forma de uma EQUAÇÃO, pois o resultado fere a condição de existência $x - 1 > 0$.

p) $4^{\log x} = 1 \Rightarrow 4^{\log x} = 4^0$

$\log x = 0 \Rightarrow 10^0 = x \Rightarrow \boxed{x = 1}$

q) $3^{2+\log_5 x} = 81 \Rightarrow 3^{2+\log_5 x} = 3^4$

$2 + \log_5 x = 4 \Rightarrow \log_5 x = 2 \Rightarrow \boxed{x = 25}$

r) $\frac{3 + \log x}{2 - \log x} = 4 \Rightarrow 3 + \log x = 8 - 4\log x$

$5\log x = 5$

$\log x = 1$

$\boxed{x = 10}$

(11) $S(t) = 1000 \cdot \log_2(1+t)$

Condição de existência: $1 + t > 0 \Rightarrow t > -1$

a) $t = 1$

$1000 \cdot \log_2(1+1) \Rightarrow$ **$S(1) = 1000$ sapatos**

b) Produção desejada: 3000 sapatos

$1000 \cdot \log_2(1+t) = 3000 \Rightarrow \log_2(1+t) = 3$

$2^3 = 1 + t \Rightarrow$

$t = 7$ anos

(12) Quantia final: $K \cdot \log_{10}(t+2) \Rightarrow 1000 \cdot \log 30$

Ananias (mês 1) $\Rightarrow 1000 \cdot \log(3)$

Ramiro (mês n) $\Rightarrow 1000 \cdot \log(n+2)$

Então: $1000 \cdot \log 30 = 1000 \cdot \log(3) + 1000 \cdot \log(n+2)$

$\log 30 = \log(3) + \log(n+2)$

$\log 30 = \log(3n+6)$

$\log\left(\frac{30}{3n+6}\right) = 0 \Rightarrow \frac{30}{3n+6} = 1 \Rightarrow 3n + 6 = 30$

$n = 8$ meses